Split-half PCA reliability

Al Xin

Jun 11, 2026

1 Data overview

The number of responsive sites for each region and age recorded are in Table 1. Within this document, recordings from the foveal confluence (FC) were omitted.

2 Procedures

2.1 Mean-centering neurons

The column means (corresponding to the mean response of a neuron averaged across all stimuli) were subtracted from each column. To prevent leakage, centering was performed separately for each split.

2.2 Split half PCA

Recordings were split into odd (training) and even (testing) repetitions of each unique image stimulus. Repetitions of the same stimulus were averaged. PCA was performed on the training set with `prcomp()`. The test set was then projected onto the PCs. The test signal was then reconstructed with $1, 2, \ldots, n$ PCs, where n was the number of responsive sites for a region/age.

I found the proportion of variance explained for the training set with cumulative sums of the PCA eigenvalues, scaled by the overall sum. For the test set, I reported the proportion of variance explained with a standard R^2 calculation.

```
# ...
x_pca_trn <- prcomp(x_splits[[1]], center = F)
trn_pcs <- x_pca_trn$rotation

tst_scores <- predict(x_pca_trn, x_splits[[2]])
tst_loadings <- tst_scores %*% t(trn_pcs)

trn_var <- x_pca_trn$sdev ^ 2
tst_rsq <- sapply(
  1:ncol(trn_pcs),
```

Table 1: Number of responsive sites found for each region and age recorded, with arrays numbered arbitrarily.

Age (wks)	Array 1	Array 2	Array 3	Total sites
V1				
29	21	NA	NA	21
37	23	0	NA	23
56	34	NA	NA	34
V2				
29	19	58	NA	77
37	21	49	NA	70
56	16	45	NA	61
V4				
29	39	3	NA	42
37	43	15	NA	58
56	41	27	NA	68
409	77	NA	NA	77
IT				
20	13	NA	NA	13
31	2	4	2	8
34	4	2	NA	6
221	11	NA	NA	11

```
  function(i) {
    tst_loadings <- tst_scores[, 1:i] %*% t(trn_pcs[, 1:i])
    data.frame(
      tst_true = matrix(x_splits[[2]], ncol = 1),
      tst_pred = matrix(tst_loadings, ncol = 1)) %>%
        lm(tst_true ~ tst_pred, .) %>%
        summary() %>%
        .$r.squared })
trn_var_prop <- cumsum(trn_var) / sum(trn_var)
# ...
```

3 Results

3.1 Different stimuli subsets and included channels

The split half PCA procedure follows the same logic as split half correlation. However, unlike correlation, blank stimuli were removed. For this analysis, I ran the split half PCA procedure with 1) all non-blank stimuli, 2) noisy stimuli, and 3) naturalistic stimuli.

I also compared performance on only responsive sites vs. all available sites. For figures with only responsive sites, the number of included PCs has been scaled to the proportion of total PCs. For figures with all available sites, I did not alter the scale of the number of PCs, but I did truncate graphs

to the first 96 PCs, corresponding to the number of channels on a single array.

The number of PCs to explain 90% of maximum test set R-squared for non-blank stimuli, noisy stimuli, and naturalistic stimuli can be found in Table 2 (in analyses with only responsive sites).The maximum test set R-squared can be found in Table 3.

3.2 R-squared mostly consistent with Lee et al.'s results

The R-squared trends in Table 2 seem consistent with Lee et al. finding that V4 and V2 have the best discriminability between noisy and naturalistic stimuli in LDA. That is, the higher reconstructed R-squareds reflect more usable information in those recordings.

3.3 IT flatter, V4 peakier

IT appears to have a flatter eigenspectrum in @fig_pca, though this might be an artifact of the low number of responsive sites/PCs. The analysis with all available plots might be helpful for region-age combinations with a small number of responsive sites. These plots show a clear plateau where the training PCs only capture noise in the test set. Instead of the 90% variance explained threshold, a better metric may be the location of the elbow where the plateau begins (not yet calculated).

The eigenspectrum of V4 seems consistently "peakier", with more weight on earlier eigenvalues. There appears to be a trend where V1 is flattest, V4 is peakiest, and V2 is in-between. Next steps can explore this more.

3.4 Worse reconstruction of noisy stimuli

In Figure 1, on the test set, naturalistic-only stimuli to have better reconstruction than non-blank, which in turn is better than noisy-only stimuli. (I double-checked that the centering of columns occurred *after* subsetting the stimuli type). This trend probably implies that the noisy texture stimuli produced more noise in the signal response.

Generally, fewer PCs are needed to explain 90% of the variance of the test set when the stimuli only include the noisy or naturalistic stimuli. Noisy-only stimuli tend to require fewer PCs than naturalistic stimuli.

3.5 Averages in early and late windows after onset

Out of curiosity, I also created Figure 2, which contains the plots of explained variance when the signal average is taken for 50 ms after onset ("early") or 100-150 ms after onset ("late"). Since averages are taken after onset time calculation, the "early" and "late" windows were different for each region (corresponding to rows in Figure 2). I did not spot any consistent or notable differences in the curves for the training set, so they are omitted for clarity.

The order of reconstruction performance, from worst to best, tends to go from early to late to full window. Eigenspectra tend to be consistent with stimulus set, regardless of the window where averages are taken.

I don't have a confident explanation as to why the slopes of the eigenspectra after the elbow are different

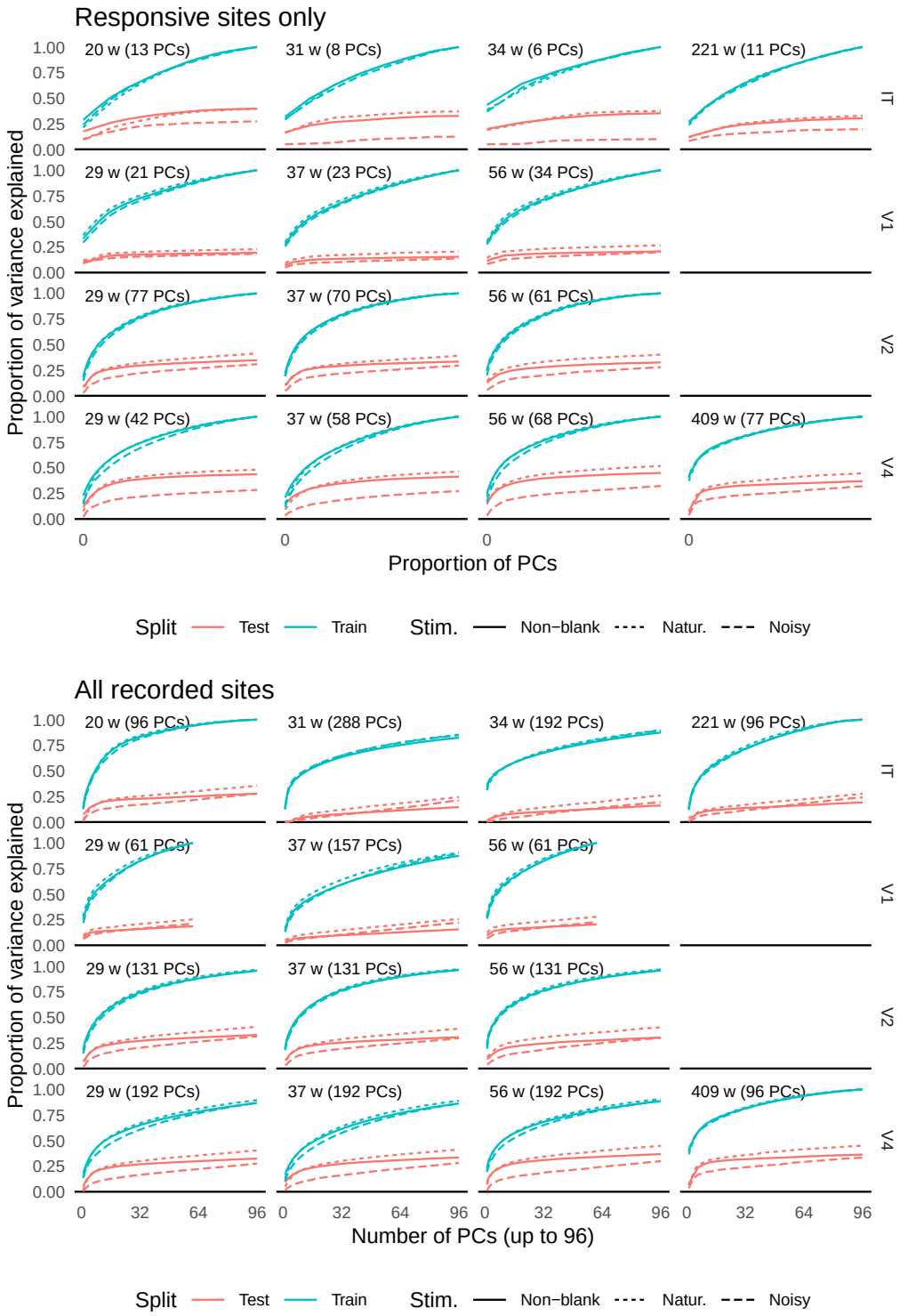


Figure 1: Comparisons between the variance explained by the principal components for the training and test set, organized by region (rows) and by session (columns). For responsive sites (top), the x-axis is rescaled for each subplot, and the true number of PCs is in parentheses. For all sites (bottom), the x-axis scale is consistent, but truncated to 96 sites.

Table 2: Number of PCs to explain 90% of the variance (scaled to proportion for test set) in responsive sites. Non-blank (NB), noisy (Ns), naturalistic (Nt).

		Train			Test		
Age (weeks)	Total PCs	NB	Nt	Ns	NB	Nt	Ns
IT							
20	13	4	4	4	6	5	6
31	8	2	2	2	4	4	2
34	6	2	2	2	3	3	3
221	11	3	3	3	5	5	5
V1							
29	21	7	8	6	15	13	9
37	23	7	7	6	12	10	7
56	34	10	11	10	23	19	11
V2							
29	77	32	34	31	40	27	20
37	70	31	33	31	42	26	19
56	61	28	30	27	33	28	17
V4							
29	42	15	16	13	27	23	14
37	58	19	20	18	31	25	16
56	68	27	28	25	42	34	19
409	77	34	36	34	48	34	23

Table 3: Maximum R-squared for split half PCA on all sites and responsive sites only. Non-blank (NB), noisy (Ns), naturalistic (Nt).

	All sites			Responsive sites		
Age (weeks)	NB	Nt	Ns	NB	Nt	Ns
IT						
20	0.278	0.355	0.278	0.398	0.396	0.273
31	0.318	0.574	0.574	0.326	0.371	0.123
34	0.252	0.426	0.396	0.352	0.374	0.099
221	0.191	0.276	0.242	0.302	0.327	0.197
V1						
29	0.185	0.253	0.212	0.193	0.228	0.185
37	0.210	0.364	0.330	0.155	0.205	0.137
56	0.204	0.277	0.227	0.207	0.267	0.197
V2						
29	0.353	0.452	0.373	0.347	0.412	0.309
37	0.332	0.441	0.360	0.334	0.392	0.296
56	0.329	0.455	0.354	0.327	0.403	0.280
V4						
29	0.394	0.540	0.440	0.437	0.482	0.284
37	0.404	0.546	0.445	0.413	0.463	0.273
56	0.434	0.576	0.459	0.449	0.518	0.322
409	0.363	0.451	0.334	0.368	0.447	0.319

for the stimulus types or even a guess as to whether it's that important. It might reflect the split-half correlation/self-consistency of neurons, though I would need to check this against the learned PCs.

4 Next steps

When noisy and naturalistic stimuli were grouped together, the eigenspectrum did not obviously differ from the eigenspectra derived from the noisy-only and naturalistic-only stimuli. This may indicate that the most influential eigenvalues/eigenvectors for all non-blank stimuli did not correspond to the noisy/naturalistic distinction. So, a reasonable next step would be to investigate predictiveness of the noisy-naturalistic distinction as a function of the number of included eigenvalues. Relatedly, checking predictiveness for other features, like the family, block order, etc. might be interesting.

Furthermore, I might want to backtrack on the responsive sites to do some additional exploratory analyses. I am somewhat interested in whether responsive sites are all up- or downregulated.

One consistent pattern throughout analyses is that V4 tends to look more "peaky"/have more influential early eigenvalues. From Lee et al.'s original publication, we know V4 is the most predictive of the noisy-naturalistic distinction. There might be something there, i.e., questions like:

- Do the initial principal components of V4 contain more information about the noisy-naturalistic distinction?
- Are the PCs of V4 linearly related to the PCs of V2? or IT?
- Does V2 lack predictiveness of the noisy-naturalistic distinction or is it nonlinearly decodable?

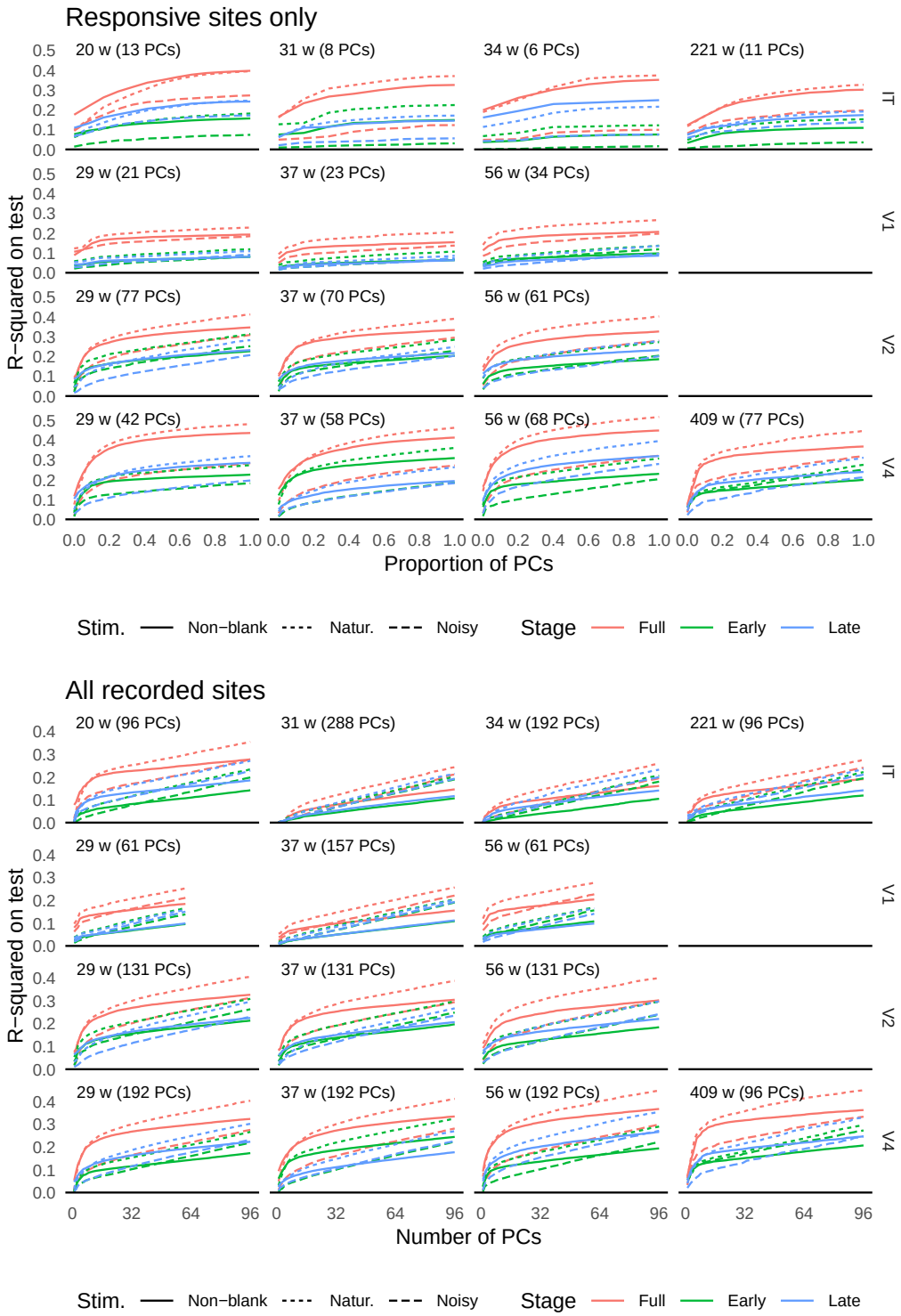


Figure 2: Comparison of proportion of variance explained for data with only responsive sites (top) and all recorded sites (bottom). Colors indicate whether averages were taken 0-50 ms after onset or 100-150 ms after onset. Training set PCA results omitted. Number of sites for each subplot is in parentheses.